

Depolarizing (Pauli) Error Channel
 For some quantum state ρ and
 physical qubit with error probability p :

$$E_{\text{pauli}}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad (1)$$

For some quantum state ρ and
 physical qubit with error probability p :

$$E_{\text{pauli}}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad (2)$$

An erasure occurs when a qubit no longer detectable by the measurement device, so the quantum state effectively “lost” or “erased”. If we extend the mathematical representation of a qubit Hilbert space with some $|e\rangle$ that is orthogonal to the qubit state $\{|0\rangle, |1\rangle\}$, then we can define the erasure channel as:

$$E_{\text{erase}}(\rho_j) = (1 - p)\rho_j + p|e\rangle\langle e| \quad (3)$$

qubit at location j leaves the computational subspace
 flagged as “erased”, we model with orthogonal state $|e\rangle$:

$$E_{\text{erase}}(\rho_j) = (1 - p)\rho_j + p|e\rangle\langle e|.$$

qubit j reset to maximally mixed state: $\rho_j = \frac{1}{2}I$.